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Impact of vaccination on SARS-CoV-2 evolution and immune escape variants

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ABSTRACT

Vaccines and host genetic factors can influence the SARS-CoV-2 evolution and emergence of new variants. Even vaccinated cases get affected as virus continues to evolve, raising concerns about vaccine efficacy and the emergence of immune escape variants. Here, we have analyzed 2295 whole-genome sequences of SARS-CoV-2 collected from vaccinated and unvaccinated cases to evaluate the impact of vaccines on virus diversity within hosts. Our comparative analysis revealed a significant higher incidence of intra-host single nucleotides variants (iSNVs) in vaccinated cases compared to unvaccinated ones (p value$<$0.0001). Furthermore, we have found that specific mutational processes, including APOBEC (C $>$ T) mediated and ADAR1 (A $>$ G) mediated mutations, were found more prevalent in vaccinated cases. Vaccinated cases exhibited higher accumulation of non-synonymous mutation than unvaccinated cases. Fixed iSNVs were predominantly located in the ORF1ab and spike genes, several key omicron defining immune escape variants S477N, Q493R, Q498R, Y505H, L452R, and N501Y were identified in the RBD domain of spike gene in vaccinated cases. Our findings suggest that vaccine plays an important role in the evolution of the virus genome. The virus genome acquires random mutations due to error-prone replication of the virus, host modification through APOBEC and ADAR1 mediated editing mechanism, and oxidative stress. These mutations become fixed in the viral population due to the selective pressure imposed by vaccination.

1. Introduction

Since the initial outbreak of severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) in December 2019, the virus has rapidly spread across continents causing a global pandemic. Like other RNA-viruses the error prone proofreading mechanism of SARS-CoV-2 has given rise to emerging variants with variable transmissibility and virulence. Based on the transmissibility and immune evasion the variants were classified into “variants of concern”, and “variants of interest” by the World Health Organization (WHO) and Centers for Disease Control (CDC). Several potential vaccine candidates came up in the early stages of SARS-CoV-2 pandemic as the primary method to control the spread of the virus. To date, 70% of the world’s population has received at least one dose of a COVID-19 vaccine [1]. All the vaccine candidates deployed globally are either inactivated whole virus, spike gene in a viral vector or mRNA-based approach. Although all the vaccination candidates have reduced disease severity for reinfection cases and reduced mortality, the highly variable nature of the virus genome has made it difficult to eradicate the pathogen.

The occurrence of SARS-CoV-2 reinfections and vaccine breakthrough infections has increased among vaccinated cases due to the continuous introduction of variations in the virus genome [2,3]. Several newly evolved, genetically distinct SARS-CoV-2 lineages have the ability to escape the vaccine induced immune response in humans [4–6]. While replicating in a host the virus accumulates mutations due to proofreading error or under the influence of host RNA editing machinery. These within-host variations, also known as intra-host single nucleotide variations(iSNVs), under the selection pressure of host immunity, give rise to newer, more potent strains and later get fixed in a population [7–9].

Sequencing based surveillance of SARS-CoV-2 has been an integral part of the disease control program and development of vaccine candidates. Beyond the determination of lineage/clade and mutational patterns, the sequencing data also provides insight into the within-host variation as the less prominent alternate alleles (AF $<$ 0.80) [10,11]. Several studies have demonstrated that iSNV dynamics is one of the early indicators of emerging immune escape strains [12,13].

In this study, we have analyzed 2295 (520 - Vaccinated, 1775 -

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Unvaccinated cases) whole genome sequences of SARS-CoV-2 from India collected between September 2020 to October 2022. In India, two vaccine candidates were developed indigenously and were primarily distributed among the population. Covaxin, an inactivated whole-virus vaccine, and Covishield, an adenovirus vector-based vaccine. The effectiveness of the Covishield vaccine (~90%) is more than the Covaxin (~80%) [10]. The majority of the vaccinated cases have administered the Covaxin and Covishield vaccines and a few individuals have taken other types of vaccine (e.g. Sputnik, BioNTech etc.). The spatiotemporal analysis of virus genomic data was studied in both vaccinated and unvaccinated cases. We explored how fixed iSNVs (SNPs) become dominant within the viral population, replacing original genetic sequences and escaping immune responses by neutralizing antibodies. The lineage diversity of SARS-CoV-2 represented by entropy was reduced in the case of unvaccinated cases in contrast to vaccinated cases.

2. Results

2.1. Patient demographics

In this study, we have analyzed 2295 (520 - Vaccinated, 1775 - Unvaccinated cases) whole genome sequence of SARS-CoV-2 samples, collected from different states of India majorly Odisha and Chhattisgarh, and a few samples from Bihar and Jharkhand between September 2020 to October 2022 (**Supplementary Fig. 1A**). The median age is 36.5 for males and 38 for females in the vaccinated cases. In contrast, the unvaccinated cases exhibit a median age of 35 for males and 30 for females (**Supplementary Fig. 1B**). The majority of the vaccinated cases have been administered the covaxin and covishield vaccines and few individuals have taken other types of vaccines (e.g., Sputnik etc.) (**Supplementary Fig. 1C**). The quality of the sequences was checked in terms of mean coverage and percentage of genome covered in both vaccinated and unvaccinated cases. Only the samples with >90% genome coverage were taken for downstream analysis (**Supplementary Fig. 1D**).

2.2. Diversity of SARS-Cov-2 lineages in vaccinated and unvaccinated cases

To understand the correlation between lineage diversity and vaccination rate in the Indian population, all India metadata was downloaded from the GISAID database (Dataset: July 2020 to Feb 2023; Count: ~2.9 L genome). Monthly diversity in SARS-CoV-2 lineages was quantified using Shannon entropy of the lineage probability distribution within 1-month time window. We observed a mixed pattern in lineage entropy distribution (Fig. 1A). Mainly, the lineage diversity increased when the new category of VOC emerged (Oct 2021; omicron evolved, May 2022; omicron (XBB)).

When we compared the lineage entropy and vaccination rates in both vaccinated and unvaccinated cases, the lineage diversity was observed higher in vaccinated cases compared to unvaccinated cases (Fig. 1B). A spike, in lineage entropy was observed in October and December of 2021 in vaccinated cases. In terms of absolute lineage count differences between vaccinated and unvaccinated cases, a total of 9 VOC lineages were observed consisting of ~98% Delta and ~ 2% Omicron in vaccinated cases, and 5 were observed in unvaccinated cases in October 2021. In December 2021, a total of 12 lineages were observed consisting of ~81% of delta and ~ 19% of omicron in vaccinated cases. Similarly, in unvaccinated cases, 9 lineages were observed consisting of ~97% of delta and ~ 3% of omicron (Fig. 1C).

2.3. Intra-host variation in vaccinated and unvaccinated cases

For each sample, iSNV genomic sites were determined, which were further filtered to identify true iSNVs using the defined threshold (see **Methods**). A total 8969 iSNVs were identified from 520 vaccinated cases and 24,415 iSNVs were identified from 1775 unvaccinated cases. Of these vaccinated iSNVs, 5281 (61.66%) of them were nonsynonymous, and 3284 (38.34%) were synonymous variants. Similarly, of these unvaccinated iSNVs, 14,278 (58.48%) were nonsynonymous and 10,138 (41.52%) were synonymous variants (**Supplementary Fig. 2A**). The number of iSNVs per sample was calculated for both vaccinated and unvaccinated cases (Fig. 2A). The mean and median number of iSNVs per sample for vaccinated cases (Median:11.5, Mean:17.52; Fig. 2A dashed line) was found to be higher compared to unvaccinated cases (Median:11, Mean:14.05; Fig. 2A dashed line). In addition, the overall percentage of iSNVs per genome was found significantly higher in vaccinated cases compared to unvaccinated ones (Fig. 2B). However, no significance was observed among the vaccinated cases between Covaxin and Covishield-administered individuals (**Supplementary Fig. 2B)**. The high incidence of iSNVs in vaccinated cases suggests that the vaccine might play an important role in SARS-CoV-2 genome evolution, which will lead to the selection of new variants that can evade immune response.

Next, we examined the iSNVs distribution across the SARS-CoV-2 genome in both vaccinated and unvaccinated cases. The number of iSNVs was found considerably higher in coding regions of vaccinated cases compared to unvaccinated cases. The average number of iSNVs was found higher in ORF1ab (Vaccinated: 10.4, Unvaccinated: 8.7) followed by the S gene (Vaccinated: 3.3, Unvaccinated: 2.6), M (Vaccinated: 2.04, Unvaccinated: 1.75) and N genes (Vaccinated: 1.91, Unvaccinated: 1.76) in both vaccinated and unvaccinated cases (Fig. 2C). There was not much difference in the mean of iSNV count were found in other genes i.e. E, ORF6, ORF3a, ORF7a, ORF8, ORF10 and untranslated regions between vaccinated and unvaccinated cases. However, when we compared the percentage of iSNVs across the SARS-CoV-2 genome between vaccinated and unvaccinated cases, the overall percentage of iSNVs was found higher in unvaccinated cases compared to vaccinated cases. Only 3`UTR, E, M, ORF6, and ORF3a genes were found to be significantly higher in unvaccinated cases compared to vaccinated ones. (p-value <0.05) (**Supplementary Fig. 2C**).

To understand the effects of immune selection of iSNVs in different VOCs of SARS-CoV-2, we compared the iSNVs distribution in different lineages for both vaccinated and unvaccinated cases. The vaccinated cases had 4 times higher iSNV frequency than the unvaccinated isolates, specifically for certain lineages. B.1 and B.1.1 lineage was found a high frequency of iSNVs in both vaccinated and unvaccinated cases followed by delta (B.1.617.2) and omicron (BA.1*, BA.2*) **(**Fig. 2D**)**. Similar to the previous report [7], we also found that VOC lineages have a higher abundance of iSNVs than the non-VOC lineages in both vaccinated and unvaccinated cases (**Supplementary Fig. 2D**). Overall data suggests that the vaccine may have specific effects on the with-in host mutation rate and VOC lineages induces more with-in host genetic variation compared to non-VOC lineages upon infection. In addition, vaccinated delta and omicron cases had significantly higher incidence of iSNVs than unvaccinated delta and omicron cases (p-value <0.0001; Fig. 2E). This may be due to different levels of immune evasion between omicron and delta, as neutralizing antibody titers induced by the Covaxin vaccine against omicron were lower than those against delta [12].

2.4. SARS-CoV-2 genome evolution driven by host factors

Understanding the evolutionary dynamics of the SARS-CoV-2 genome is essential for monitoring the emergence of new variants and assessing their potential impact on public health. SARS-CoV-2 undergoes evolutionary changes over time by accumulating mutations through various mechanisms, including errors made by the viral replication machinery or exposure to the host immune response [13]. It has been reported that the SARS-CoV-2 genome can be edited by cellular mutational drivers like apolipoprotein B mRNA-editing catalytic polypeptide-like deaminase (APOBEC) and its family and different cellular antiviral proteins like adenosine deaminase RNA 1 (ADAR1) [14–17]. APOBEC family proteins deaminase cytosine to uracil (C > U or

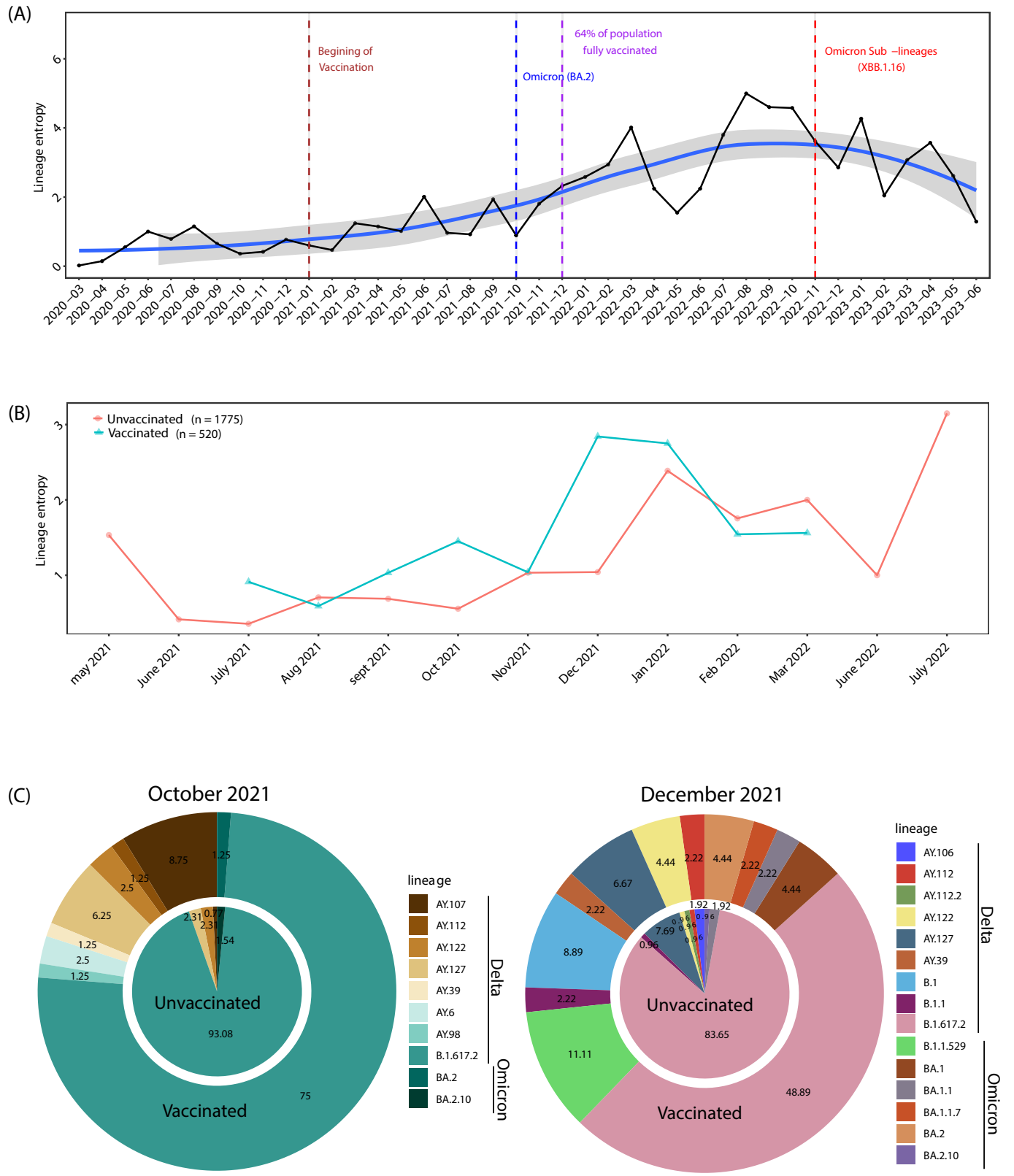

Fig. 1. Diversity of SARS-Cov-2 lineage in vaccinated and Unvaccinated cases:
A. SARS-CoV-2 lineage diversity index (Shannon entropy) among the Indian samples. The vertical dashed line indicates the vaccination event and the emergence of sub-lineages during that time. The blue line indicates the mean, and the grey shaded area indicates the standard deviation.
B. SARS-CoV-2 lineage diversity index (Shannon entropy) among the vaccinated and unvaccinated cases across time.
C. The proportion of Delta and Omicron lineages observed in vaccine breakthrough and unvaccinated infections after >70% of the population received at least two vaccine doses. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

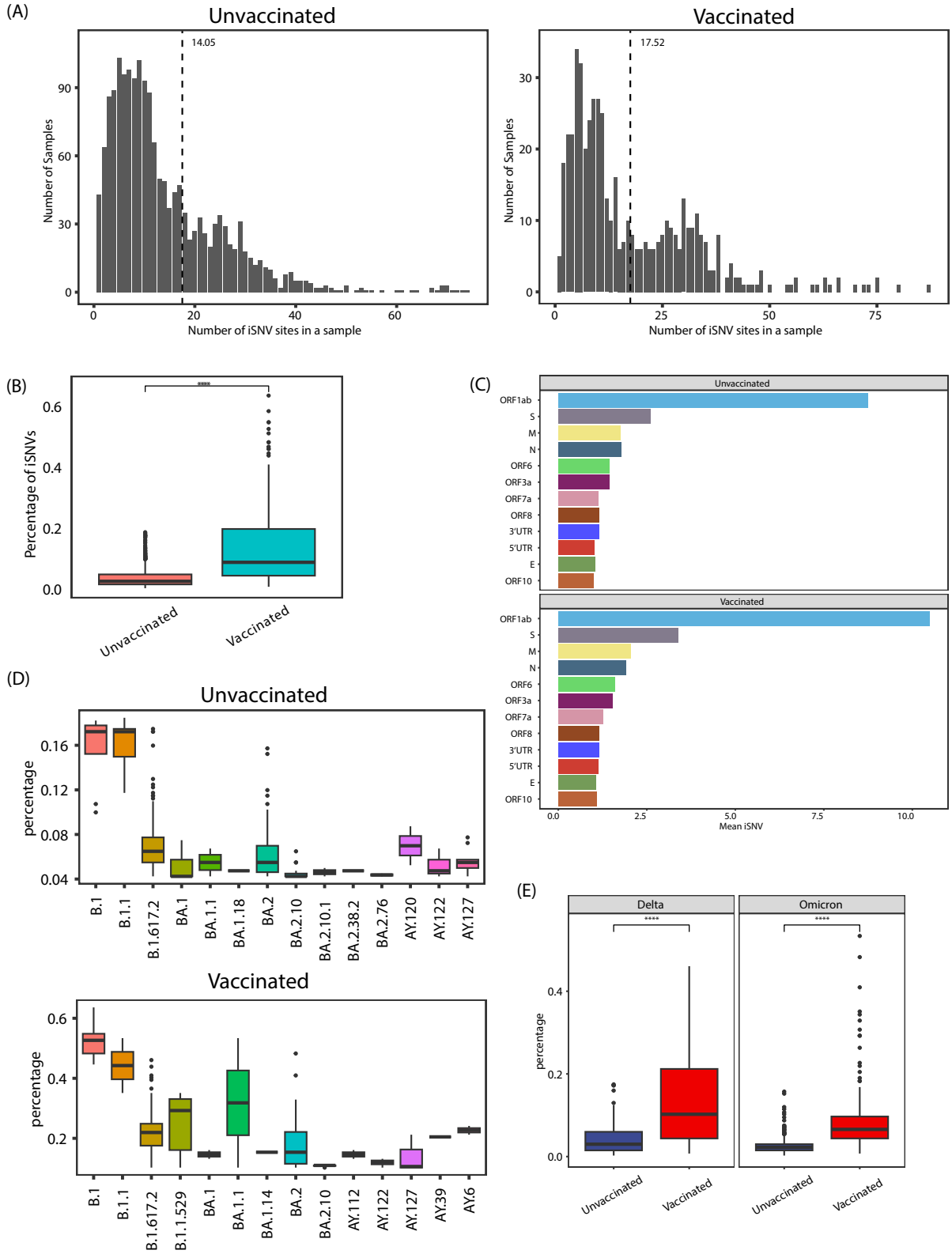

Fig. 2. Intra-host variation in vaccinated and unvaccinated cases:
A. Distribution of number of iSNVs in vaccinated and unvaccinated cases. The dashed line shows the mean value of the distribution.
B. Comparison of iSNV observed in vaccine breakthrough infections in contrast to the unvaccinated infections.
C. Mean iSNVs distribution in coding and un-translated regions of SARS-CoV-2 genome.
D. Distribution of incidence of iSNVs in different lineages of SARS-CoV-2.
E. Comparison of percentage of iSNVs in delta and omicron lineages between vaccinated and unvaccinated cases.

C > T and G > A) and ADAR1 proteins deaminase adenosine into inosines (A > I or A > G and U > C or T > C) [14]. Evidence suggests that even reactive oxygen species (ROS) induce oxidative DNA damage and convert guanine to oxoguanine which can pair with adenine and lead to G > T substitution [17].

We checked the iSNV frequency (normalized iSNV per kb) of synonymous and nonsynonymous mutations in all mutational types in both vaccinated and unvaccinated cases. Our analysis revealed that across all mutational types, vaccinated cases exhibited a higher prevalence of iSNVs than unvaccinated ones. Consistent with previous findings, the majority of mutations were mediated by APOBEC (C > T) [18], followed by ROS (G > T) [19], and ADAR1 mediated mutation (A > G and T > C) [20] in both vaccinated and unvaccinated cases (Fig. 3A). Then we compared the incidence of iSNVs between vaccinated and unvaccinated cases for each mutational type. APOBEC and ADAR1 mediated mutations (C > T and A > G) were significantly more prevalent in vaccinated

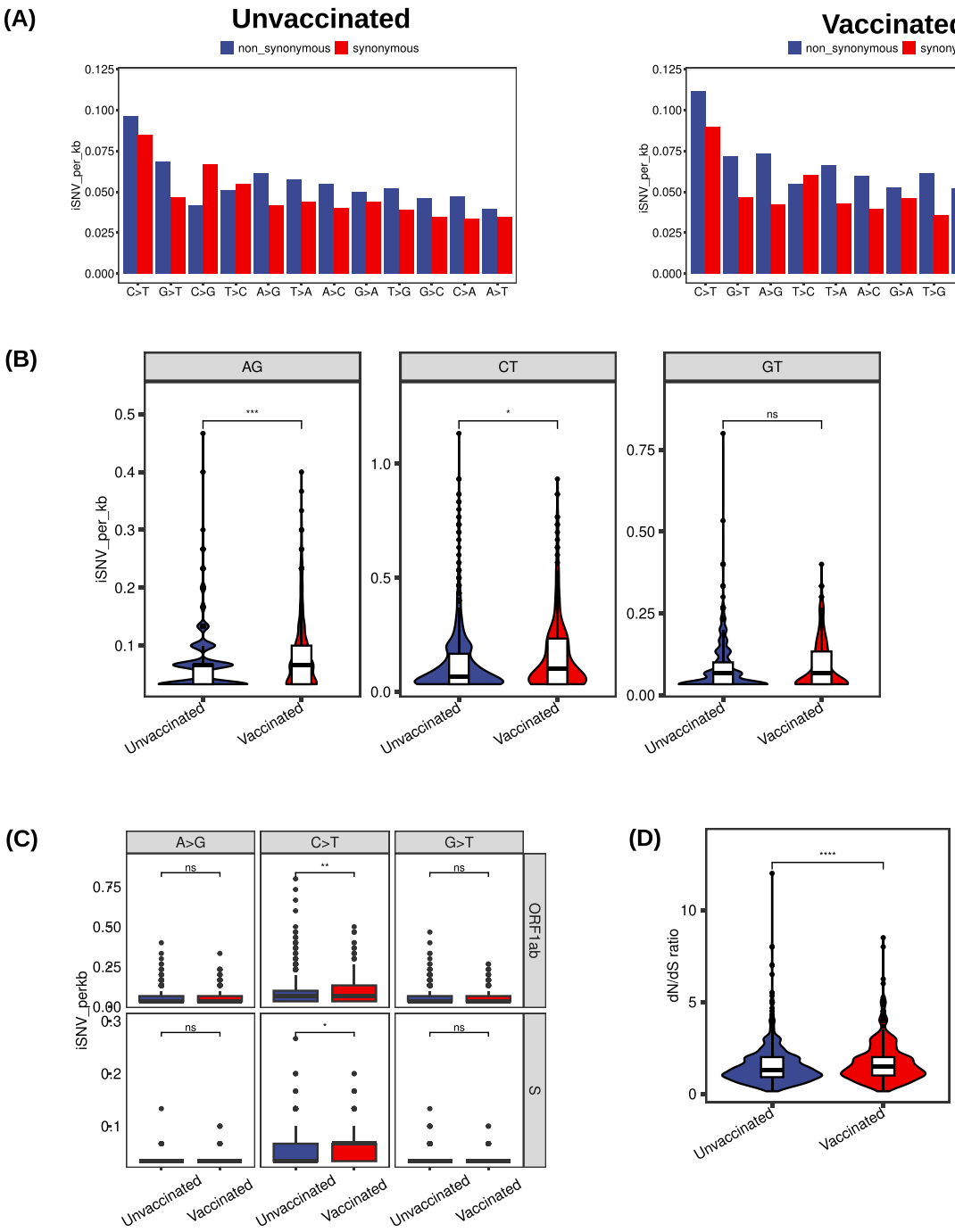

Fig. 3. SARS-CoV-2 genome evolution driven by host factors:
A. Normalized iSNV number for different mutation types. Synonymous (red) and non-synonymous (dark blue) mutations are distinguished by color.
B. Comparison of normalized iSNV frequencies for C > T, G > T and A > G mutation types in vaccinated and unvaccinated cases.
C. Comparison of normalized iSNV frequencies among C > T, G > T and A > G mutation types within two specific genes, ORF1ab and S, in samples from both vaccinated and unvaccinated cases.
D. Non-synonymous/synonymous (dN/dS ratio) of identified iSNVs between vaccinated and unvaccinated cases. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

cases. Interestingly, despite being the second-highest mutational type in the population, we did not observe any significant difference in the incidence of ROS-mediated mutation G > T between vaccinated and unvaccinated cases (Fig. 3B). Other mutation types were also found significantly high in vaccinated cases except for the A > C, C > A, and G > C mutation types (**Supplementary Fig. 3A**). To identify which genes are prone to accumulating variants we compared the incidence of iSNVs for the top mutational types (C > T, A > G, and G > T) between vaccinated and unvaccinated cases for each gene. For C > T (APOBEC mediated), both ORF1ab and S genes exhibited statistical significance (Fig. 3C). While A > G (ADAR1 mediated), only the ORF3a gene showed significance (**Supplementary Fig. 3B**). These results suggest that RNA editing mediated by APOBEC and ADAR1 are one type of possible mechanisms for SARS-CoV-2 genome diversity but other mechanisms also can be explored.

Further, to test whether the observed mutation was caused by fitness selection in vaccinated cases, we estimated the selection pressure by calculating the dN/dS ratio, where dN represents the rate of non-synonymous substitutions and dS denotes the rate of synonymous substitutions. We observed a significantly higher abundance of non-synonymous mutations in vaccinated cases compared to unvaccinated ones, indicating potential fitness-driven selection (p value<0.0001) (Fig. 3D).

2.5. Temporal trajectory of within-host variations

SNPs were identified in vaccinated and unvaccinated cases to study the variants fixed from iSNVs to SNPs. Then each genomic site of iSNVs was compared with the SNPs. After removing single iSNV genomic sites that appeared once in the population, out of 1012 iSNV genomic sites, 250 (~ 24%) of iSNVs were found fixed in vaccinated cases, and out of 1896 iSNVs genomic sites, 475 (~ 25%) iSNVs were fixed in unvaccinated cases, (Fig. 4A). When we checked the distribution of fixed iSNVs in different genomic regions of SARS-CoV-2, ORF1ab and S genes were found to be the top genes in the number of fixed mutations in both vaccinated and unvaccinated cases (**Supplementary Fig. 4A**). Then we compared the frequency of these fixed iSNVs genomic site variants between vaccinated and unvaccinated cases over time. Three mutational types (A > C, G > C, and T > A) were observed with initial allele frequencies within the range of 0.2 to 0.3. These mutations underwent fixation over time within the population, with the allele frequencies ranging from 0.8 to 1.0 (Fig. 4B). For A > C and T > A mutation types, we observed that the alternative alleles achieved fixation earlier in the case of vaccinated cases, and after a three-month interval, fixation occurred in unvaccinated cases. However, for G > C mutation type, we did not find any significant difference in fixation time between vaccinated and unvaccinated cases (Fig. 4B). When we compared the frequency of iSNVs (normalized iSNVs per kb) of these three mutation types (A > C, G > C, and T > A) in coding regions of SARS-CoV-2, ORF1ab gene was found significant for T > A and S gene was found significant for A > C mutation type (**Supplementary Fig. 4B)**.

2.6. Emerging immune escape variations in vaccine breakthrough cases

As vaccines increase the incorporation of new mutations into the SARS-CoV-2 genome, all fixed iSNVs from both vaccinated and unvaccinated cases were compared with curated immune escape-variants in the SARS-CoV-2 ESC database [21]. A total of 17 (6.8%) immune escape variant sites were found out of 250 fixed iSNVs in vaccinated cases and 18 (3.78%) sites were found among 475 fixed iSNVs in unvaccinated cases. (Fig. 4A). Overlapping the immune escape variant among both vaccinated and unvaccinated fixed variants revealed 11 common immune escape variants, 6 immune escape variants that were unique to the vaccinated cases, and 7 immune escape variants unique to the unvaccinated cases (Fig. 4C). A large proportion of the immune escape variants were located in the receptor binding domain (RBD) of spike protein (S: c.22515–23,185), which plays a key role in binding to the host angiotensin-converting enzyme 2 (ACE2) receptor of the host cell. A majority of the immune escape variants in spike protein are Omicron lineage-defining changes i.e. E484A, N501Y, Q493K, K417N, S477N, Y505H, and G496S. The prevalence of Omicron was found highest (22.1%) among the vaccinated cases compared to the unvaccinated cases (10.7%) (**Supplementary Fig. 4C)**.

3. Discussion

Since the outbreak of the first case of COVID-19, several thousands of mutations including SNPs have been identified. Some of these mutations may lead to changes in the virus that allow it to escape the immune response elicited by vaccines or previous infections. In this study, we have analyzed 2295 SARS-CoV-2 samples to find out the impact of vaccines in terms of incidence of intra-host variations and fixing in the population by doing a comparative analysis between vaccinated and unvaccinated cases. Significance differences in intra-host variation in SARS-CoV-2 between vaccinated and unvaccinated cases suggest that vaccines play an important role in the SARS-CoV-2 evolution and selection processes.

Our comparative analysis provided evidence that immune selection shapes the evolution of the SARS-CoV-2 genome and helps in the accumulation of escape variants. Previous reports have suggested that SARS-CoV-2 lineage diversity is correlated with the vaccination rate (Global Dataset: Dec 2020 to May 2021; Count: 1.8 M genome) [22]. Despite the no correlation between vaccination rate and lineage entropy, we observed a high abundance of iSNV sites among the vaccinated cases (mean iSNV per sample: 17.52). Our study found that the Delta and Omicron variants were the contributing lineages for the higher occurrence of iSNV sites in vaccinated cases. The observation suggests that there may be a probable link between the emergence of VOCs and with increased number of immune escape variations in late 2021.

Despite the no correlation between vaccination rate and lineage entropy in our samples, we observed that compared with unvaccinated cases, vaccinated cases (mean iSNV per sample: 17.52) showed high incidence of iSNVs, suggesting that vaccines impact the evolution of SARS-CoV-2 genome. Different lineages of SARS-CoV-2 can exhibit distinct properties, including variations in transmissibility [23,24], disease severity [25,26], viral load [26,27], tissue affinity [28] and vaccine breakthrough [4,6,29]. Here, we found that delta and omicron from vaccinated cases showed high iSNVs rate compared to unvaccinated cases. This suggests that vaccination may influence the genetic diversity and evolution of the virus within the host, leading to the emergence of genetic changes that allow the virus to evade or partially escape the immune response. We also found significant differences within host mutation rates between VOC and non-VOC lineages of SARS-CoV-2, suggesting differences in intrinsic biological properties among lineages.

Additionally, the evidence of RNA editing at genome level was observed in our study. Similar to the previous report, we also observe that APOBEC mediated mutation (C > T) followed by ROS (G > T) and ADAR1 (A > G) mediated mutation type had high frequency [7,17,30] and APOBEC and ADAR1 mediated mutation types were found significantly high in vaccinated cases, which suggest that these process mediated RNA editing might be prevalent in the vaccinated cases. Report suggests that ADAR1 mediated RNA editing of the viral genome leads to autoimmunity but favors viral infection, eventually it helps viruses to evade immune response [31,32]. The frequency of non-synonymous to synonymous (dN/dS ratio) mutations ratio was found higher in vaccinated cases than the unvaccinated cases. The fixed iSNVs (SNPs) were majorly distributed in the ORF1ab and S gene and three mutation types (A > C, G > C, T > A) became stable overtime in vaccinated cases. Early selection of these mutations in vaccinated cases compared to unvaccinated cases suggest that selection of these

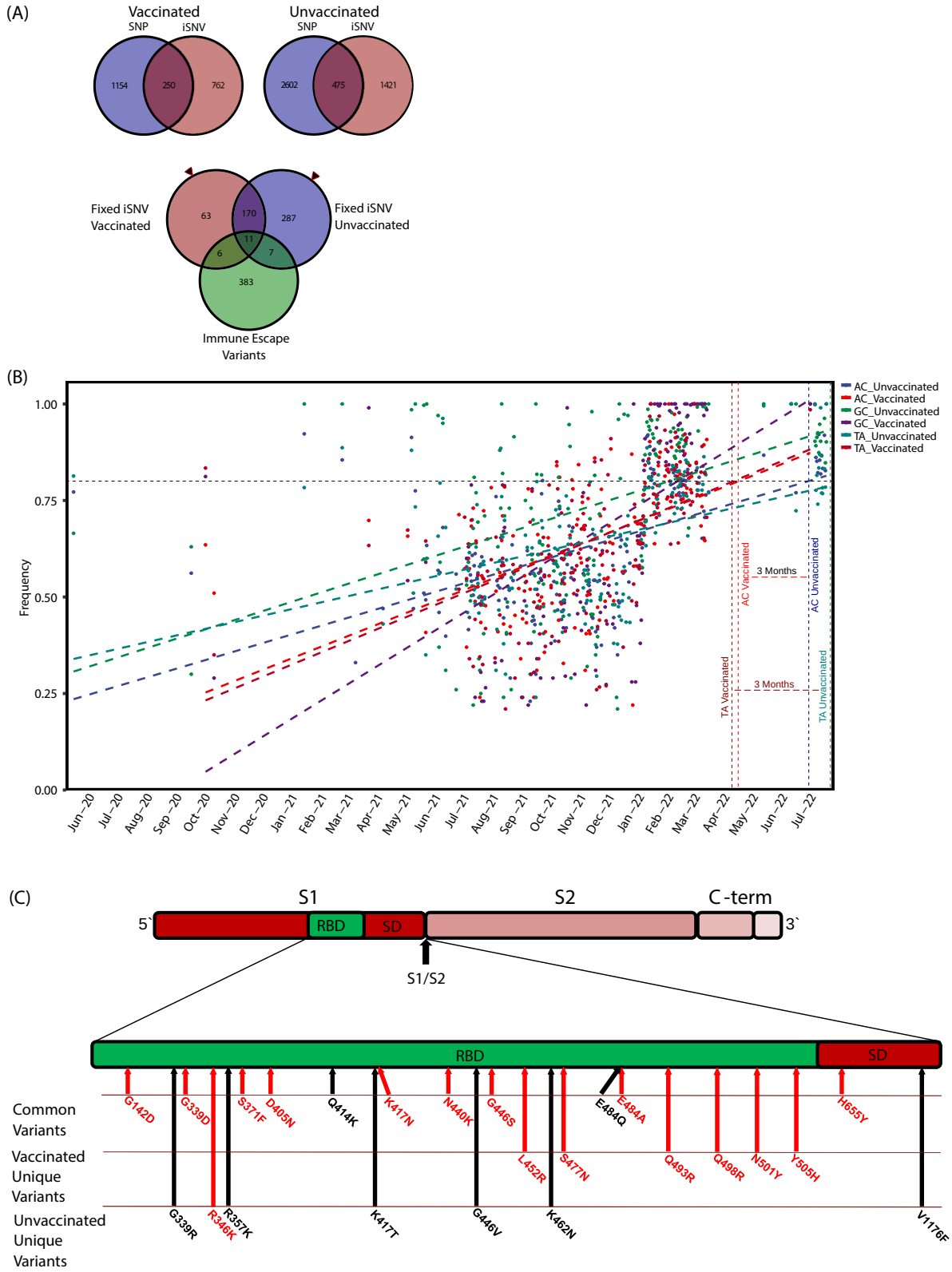

Fig. 4. Emerging immune escape variations in vaccine break through cases.

A. Number of fixed variants in vaccinated and unvaccinated groups. Common and unique immune escape variants in vaccinated and unvaccinated groups.

B. Occurrence of three fixed iSNV alleles A > C, G > C and T > A across the time in vaccinated and unvaccinated samples. A > C and T > A achieved selection three months earlier in the vaccinated group compared to unvaccinated group.

C. Location of the common and unique immune escape variants detected in SARS-CoV-2 infection groups in the Spike protein. Red color indicates omicron specific mutations and black color indicates other mutations. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

mutations is driven by vaccines. Collectively, vaccines have an important role in SARS-CoV-2 evolution and its viral diversity. It also suggests that the immune pressure exerted by vaccination might be driving the selection of specific viral variants, potentially favoring mutations that escape the immune response.

Evidence suggests that vaccination induces selective pressure that facilitates the rise of new mutations and that might promote immune escape mechanisms [33,34] and earlier study shows that omicron variants are more prone to immune escape than the delta variants in terms of energy and conformational perspectives [35]. So we checked the immune escape variants from fixed iSNVs (SNPs) of both vaccinated and unvaccinated cases and several common and unique omicron specific key mutations were found in the receptor binding domain (RBD) of the spike protein of the genome. Four omicron specific variants: S477N, Q493R, Q498R, Y505H, one delta and omicron shared common variants L452R and one alpha, beta, gamma and omicron shared common variants N501Y, total six unique immune escape variants were found in vaccinated cases and eleven common immune escape variants: G142D, G339D, S371F, D405N, Q414K, K417N, N440K, G446S, E484Q, E484A and H655Y were found in both vaccinated and unvaccinated cases. Earlier reports suggest that these key mutations of omicron provide virus immune escape or high affinity towards the receptor [33,36]. The E484A, N501Y, K417N, and Y505H mutations in the RBD region of the spike protein increases infectivity by interacting with the ACE2 receptor with high affinity [37].

In conclusion, our study highlights the complex interplay between vaccination, viral evolution, and the emergence of immune escape variants. Understanding these dynamics is crucial to control the spread of COVID-19 and manage potential future outbreaks. Continued genomic surveillance and monitoring viral evolution are essential to make vaccination strategies more effective against emerging variants.

3.1. Limitation of the study

Our study compares the dynamic changes of iSNVs between vaccinated and unvaccinated cases. However, due to limitations of individual's metadata information (e.g. CT value, severity, antibody titer value etc.), we were unable to incorporate this information into our analysis.

4. Materials and methods

4.1. Sample collection and isolation

All samples were collected from SARS-COV-2 infected patient's nasal, nasopharyngeal and oropharyngeal swabs and kept in 3 mL sterile viral transport medium (VTM) tube or 1 mL of TRIzol reagent. All samples were stored at a cold temperature (2–8 °C) for short-term and stored at −80 °C for long-term use. RNA extraction was done using the TRIzol method. In order to detect SARS-CoV-2 viral infection, Real-time PCR assay was performed. This one step assay was done with the help of Q-line Molecular, Coronavirus (COVID-19) RT-PCR kit targeting CoV-2 explicit ORF1ab and N gene on QuantStudio 6 Flex System. The manufacturer's protocol was followed precisely.

4.2. Library preparation and sequencing

The sequencing libraries for this study were prepared using the Illumina COVIDSeq protocol, followed by standard protocols. The synthesized cDNA was then amplified using a multiplex polymerase chain reaction (PCR) protocol, resulting in 109 amplicons covering the SARS-CoV-2 genome (https://github.com/artic-network/primer-schemes/tree/master/nCoV-2019/V3). The PCR products underwent tagmentation and adapter ligation using IDT for Illumina Nextera UD indexes Set A, B, C, and D, which includes a total of 384 indexes for 384 cases. Further, enrichment and cleanup steps were performed as per the protocol provided by the manufacturer (Illumina). The cases were processed in batches in 96-well plates, each plate consisting of one COVIDSeq positive control HT (CPC HT) and a no template control (NTC). The final sequencing was performed as dual-indexed paired-end sequencing with a read length of 36 base pairs on the NextSeq 550 platform.

4.3. Data analysis

4.3.1. Quality control

To reduce false positive or error rate, quality of the raw fastq files were checked using FastQc tool (v0.119) [38]. Adapter and low quality bases at the end of the reads were trimmed using Cutadapt tool (v4.1) for downstream analysis [39].

4.4. Alignment and variant calling

All paired-end reads were mapped against the hg38 human reference genome. All unmapped reads were extracted using samtools [40] and mapped against SARS-CoV-2 reference genome (NC_045512.2) using BWA mem tool (v0.7.17) (https://github.com/lh3/bwa). All PCR duplicate reads were removed to avoid the false positive rate of reads using picard tools (https://github.com/broadinstitute/picard). The consensus FASTA files were generated and SNPs were called for each sample using BCFtools [39] and the SNPs were annotated using SnpEff (v5.1) and SnpSift (v5.1) [41].

4.5. iSNVs detection

Intra-host single nucleotide variations are single nucleotide variants in which an alternative allele co-existed with the reference allele at the same genomic position within the same sample. SNPs are single nucleotide variants in which the alternative allele completely replaces the reference allele at the same genomic position and must be present in at least 1% of the population.

Fixed iSNVs are the single nucleotide variants present in the virus genome during the early phase of the pandemic, and later they get fixed at the population level as SNPs due to selection pressure.

Intra-host single nucleotide variations were called from individual bam files using REDitools2 [42]. Different parameters were used to filter the iSNVs. The iSNVs sites were kept with frequency ranges from 0.2 to 0.8 for further analysis. Further, mean quality score < 30 and coverage $q30 < 5$ were removed and remaining iSNVs were considered for further analysis.

4.6. Statistical analysis

To quantify the genetic diversity of the SARS-CoV-2 genome over time, we have calculated the lineage entropy of each pangolin-defined lineage, for a given time *t*.

The lineage entropy was defined as,

$$\text{Lineage entropy} = -\sum_{1\rightarrow \text{all lineages}} \frac{\text{Number of sequences of lineage (t)}}{\text{Total number of sequences (t)}} \log\left[\frac{\text{Number of sequences of lineage (t)}}{\text{Total number of sequences (t)}}\right]$$

4.7. Data handling and visualization

Custom R and bash scripts were used for data analysis as well as to process samples through the pipeline. R libraries such as dplyr (https://github.com/tidyverse/dplyr), tidyr (https://github.com/tidyverse/tidyr) and ggplot (https://github.com/tidyverse/ggplot2) for data handling and visualization. Pairwise two-sided *t*-test was used to calculate the statistical significance.

Supplementary data to this article can be found online at https://doi.org/10.1016/j.vaccine.2024.07.054.

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Institutional review board approval

The study plan was approved by the Institute of Life Sciences Institutional Biosafety Committee (Ref no. V-122-MISC/2007–08/01) and the Institutional Human Ethical Committee (Ref no. 109/HEC/21). All necessary patient/participant consents have been obtained, and the appropriate institutional forms have been archived.

CRediT authorship contribution statement

Deepak Jena: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Data curation, Conceptualization. **Arup Ghosh:** Writing – review & editing, Supervision, Methodology, Investigation, Conceptualization. **Atimukta Jha:** Resources. **Punit Prasad:** Writing – review & editing. **Sunil Kumar Raghav:** Writing – review & editing, Supervision, Resources, Funding acquisition.

Declaration of competing interest

The authors declared no competing interests.

Data availability

The assemblies of the SARS-CoV-2 genomes have been uploaded to the GISAID database and raw data accession numbers are included in the supplementary data file (Supplementary data).

All raw data and metadata have been submitted to the IBDC portal (https://india.rcb.ac.in/insacog).

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